

Next CW Trainer

User Manual

App version 1.0.0 · September 2026 · c6b71c7

Next CW Trainer – User Manual, for app version 1.0.0 (source c6b71c7), September 2026.

Next CW Trainer is an independent project. Its training concept and algorithms come from the Morserino-32 firmware by Willi Kraml, OE1WKL, and the Morserino-32 team; there is no other connection to them.

This manual is also available in German.

Contents

1	Introduction.....	5
	Where the app comes from	5
	About this manual	5
2	Getting started.....	6
2.1	Installation.....	6
2.2	The home screen	6
2.3	Controls inside the trainings	7
2.4	Light, dark, language	7
2.5	Sound and volume	7
3	Basics.....	8
3.1	Speed in WPM	8
3.2	Spacing in dits: Interchar Spc and InterWord Spc	8
3.3	Effective speed.....	8
3.4	The Koch method	8
3.5	Prosigns.....	9
4	Listen: practise copying	10
4.1	Choosing the character set and content.....	10
4.2	Speed.....	11
4.3	How a block runs.....	11
4.4	The result page.....	13
4.5	Settings in the * sheet (Listen)	14
5	Send: Echo Trainer.....	16
5.1	Start view.....	16
5.2	How a word runs.....	17
5.3	Answer speed	18
5.4	The result page.....	18
5.5	Settings in the * sheet (Send).....	19
5.6	Practising a single character.....	20
6	Adaptive mode.....	21
6.1	What the app tracks.....	21
6.2	The two thresholds	22
6.3	When the next Koch character comes	22
6.4	Pauses and speed	23
6.5	Accepting, rejecting, adjusting suggestions.....	23
6.6	Weak characters	23
6.7	Progress card “On the way to X”	24
6.8	Trend.....	24
6.9	Adaptive mode settings.....	25
7	Character statistics	26
8	CW Keyer	27

9 CW Decoder	28
9.1 Operation	28
9.2 Decoder settings	29
10 WiFi Trx	30
10.1 Services	30
10.2 Sending and receiving	30
11 QSO Bot	32
11.1 Starting	32
11.2 QSO types	32
11.3 Talking to the bot	32
11.4 QSO Bot settings	33
12 Games	34
12.1 Morsel	34
12.2 Morse Invaders	35
12.3 Memory Chain	36
13 Paddle and Morse key	37
13.1 Touch paddles	37
13.2 A real paddle or straight key	37
14 Settings	39
14.1 Appearance	39
14.2 General	39
14.3 Keyer	40
14.4 Audio output	40
14.5 Call Signs	40
14.6 vband Paddle and Analyze Key Events	41
14.7 Info: version and build	41
14.8 Koch Sequence	41
15 What the app does not do (yet)	43
16 Troubleshooting	44

1 Introduction

Next CW Trainer is an Android app for learning and practising Morse code (CW). It brings the training modes of the Morserino-32 to your phone or tablet: Koch method, Echo Trainer, CW Keyer, CW Decoder, WiFi Trx, QSO Bot and several games. On top of that it adds an **adaptive block flow**. The app tracks your mistakes character by character and suggests when you are ready for a new character, more speed or shorter pauses.

Where the app comes from

Next CW Trainer is an independent hobby project. It has **no connection** to Willi Kraml (OE1WKL) or the Morserino-32 team. The only link is that its algorithms and training logic were read out of the Morserino firmware source and rewritten for Android. The idea behind the Morserino-32 and its training concept are Willi's and his team's work, and so are years of careful refinement. That includes the Koch sequences, the Echo Trainer, the QSO Bot and much more. The app contains no original firmware code and is not an official product of the Morserino project.

If you don't own a Morserino-32 yet, build or buy one. It's a wonderful device. This app is for practising on the go, not a replacement.

The app is based on firmware version 9.0.0. Later firmware changes are not carried over automatically.

About this manual

This manual describes the app version printed on the title page. You can see which version you have installed under **Settings → Info** (see Info: version and build).

Terms taken over from the firmware are kept as they are in the app, for example **Interchar Spc**, **Random Groups** and **Echo Prompt**. That way you can find them again in the Morserino manual.

2 Getting started

2.1 Installation

The app is currently handed out as an APK file, not through the Play Store.

1. Copy the APK file to your phone, for example by messenger, e-mail or USB cable.
2. Tap the file in your file manager. The first time, Android asks whether the app you install from may **install unknown apps**. That might be your file manager or browser. Allow it for that app.
3. Confirm the installation.

Updates work the same way: install the new APK over the old one, and your settings and statistics are kept. This only works if both APKs are signed with the same key. If Android reports a conflict, you first have to uninstall the old version, and that deletes your settings and statistics.

2.2 The home screen

The home screen has three groups:

Group	Tile	What for
Practice	Listen	Practise copying: CW Generator and Koch Trainer in the block flow
	Send	Practise sending: Echo Trainer. A word is played and you key it back
Free	CW Keyer	Key freely, with the text decoded on screen
	CW Decoder	Copy CW through the microphone
	WiFi Trx	CW over the internet with other Morserinos and apps
	QSO Bot	A simulated QSO partner
Play	Games	Morsel, Morse Invaders, Memory Chain

The **Listen** and **Send** tiles show your current Koch lesson and speed. **Send** also shows the trend of your last blocks (see Trend).

The gear icon at the top right opens the **global settings** (chapter Settings). Anything that concerns only one training is set up directly inside that training.

2.3 Controls inside the trainings

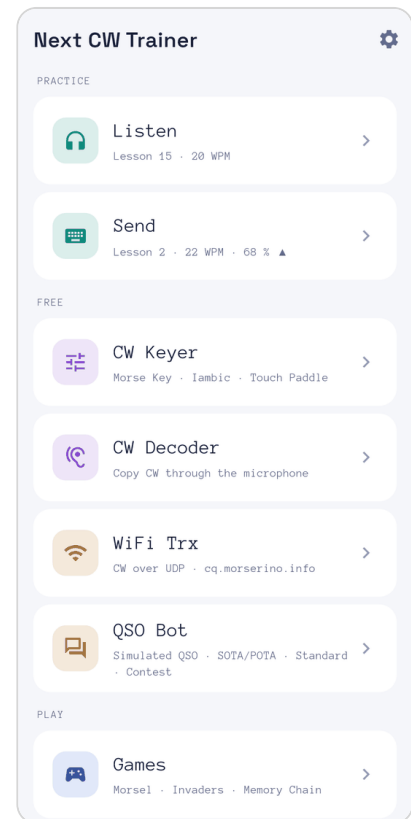
Each training has these icons at the top right:

-  **Training settings** opens a sheet that slides up from the bottom. Every change takes effect immediately and is saved.
-  **Statistics** (only in **Listen** and **Send**) shows your progress per character (see Character statistics).

While a block is running, these icons and the selection at the top are hidden to keep the screen calm. The **back button** then only ends the running block and returns you to the training's start view. A second back leaves the training.

The screen stays on while you practise.

Text size: you can **pinch** text areas that show decoded text or results to make the text larger or smaller. This works in the CW Keyer, in Send and on the result page, among others. Each area remembers its own size.



The home screen

2.4 Light, dark, language

Under **Settings** → **Appearance** you choose the **Theme** (System, Light or Dark) and the app **Language** (Deutsch or English). Both take effect immediately.

2.5 Sound and volume

Set the volume with the phone's volume keys. You set the **pitch** and the **tone softness** in the settings. When headphones, a USB audio device or a Bluetooth device is connected or disconnected, the app switches to it automatically. If you don't want that, you can fix the output (see Audio output).

Bluetooth headphones usually add noticeable latency. That doesn't matter for listening, but it does for sending: you hear your sidetone noticeably later than you key. For sending, use wired headphones or the speaker.

3 Basics

3.1 Speed in WPM

Speed is given in **WPM** (words per minute), based on the standard word “PARIS”. At w WPM a dit lasts exactly $1200 / w$ milliseconds, so 60 ms at 20 WPM.

3.2 Spacing in dits: Interchar Spc and InterWord Spc

As on the Morserino, the pauses are set **in dit lengths**:

- **Interchar Spc** is the pause between two characters of a word. Normal Morse is 3 dits. You can set 3 to 45.
- **InterWord Spc** is the pause between two words or groups. Normal Morse is 7 dits. You can set 6 to 105.

The pause between the elements (dits and dahs) *inside* a character is always 1 dit. So the characters themselves always come at the set speed, and only the pauses between them get longer. This is the **Farnsworth method**. You learn the sound of a character at full speed and still have time to think.

Next to each spacing slider the app also shows the pause **in seconds** at the current speed, for example “28 dits · 1.68 s @ 20 WPM”.

The **Listen** and **Send** trainings start with generous pauses of **28 / 40 dits**. The CW Keyer, WiFi Trx and the QSO Bot use the normal 7 dits as word spacing.

3.3 Effective speed

Because the pauses are longer, whole words arrive more slowly. The app shows this as **eff.** (effective WPM):

$$\text{eff. WPM} = 50 \times \text{WPM} / (31 + 4 \times \text{Interchar Spc} + \text{InterWord Spc})$$

Example: 20 WPM with 28/40 dits gives $50 \times 20 / (31 + 112 + 40) \approx 5$ WPM. The characters sound like 20 WPM, but you have as much time as at 5 WPM. With 3/7 dits the two values are the same.

3.4 The Koch method

With the Koch method you start with **two characters** at full speed. Once you recognise them reliably, the next character is added, then the next, until you know them all. Each step is called a **lesson**. The lesson number is the number of active characters.

The order of the characters is set by the **Koch Sequence** (see Koch Sequence).

When drawing random characters, the app weights them like the Morserino. Two out of three characters are drawn evenly from **all** active characters. Every third one is drawn only from the **last third**, where the most recently learned characters are. New characters therefore come up more often, without the old ones disappearing.

3.5 Prosigns

Prosigns (procedural signals) are sent as run-together letters. The app writes them in angle brackets:

Prosign	Meaning
<ka>	Start of message
<ar>	End of message (+)
<kn>	Only the station called should answer
<sk>	End of contact
<as>	Wait
<ve>	Understood
<bk>	Break, over to the other station
<err>	Error: eight dits (.)

In Listen (block flow), prosigns from **Random Groups** are shown and graded as two separate letters: <ka> appears as “K A”.

4 Listen: practise copying

Listen is the app's CW Generator and Koch Trainer. The app plays a **block** of groups or words, and you copy on paper. Nothing is shown on screen while it plays. Afterwards the app reveals the text and you tap what you got wrong. You then get a result with suggestions for the next block.

4.1 Choosing the character set and content

At the top of the start view you choose two things.

The **character set** decides which characters are practised:

Character set	Meaning
Koch lesson	Only the characters up to your current lesson
All characters	All letters, digits, punctuation and prosigns
Practice set	Only the characters you enter yourself

The **content** decides what is played:

Content	Meaning	Available with
Random	Groups of random characters	all
Words	Common English words	Koch, All
Abbrevs	Common CW abbreviations	Koch, All
Callsigns	Random, realistic call signs	All
Mixed	Words, abbreviations and groups mixed	Koch, All

With the Koch lesson, only words and abbreviations made **entirely** of characters you have already learned are used. In the first lessons there are very few of them.

4.1.1 Setting the Koch lesson

The **KOCH** slider sets the lesson. Below it you see all characters active in that lesson. Their colour shows the kind of character: letters, digits and punctuation/prosigns are coloured differently.

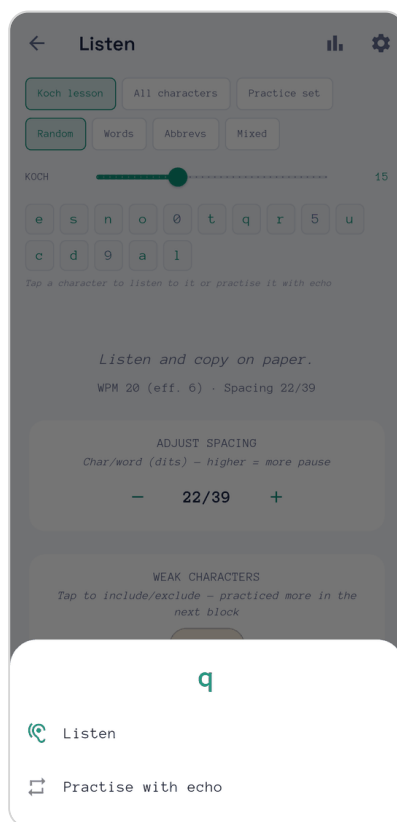
Tap a character to get to know it:

- **Listen** plays the character three times at the current speed.
- **Practise with echo** opens a single-character drill. The character plays over and over and you key it back with the paddle (see Practising a single character).

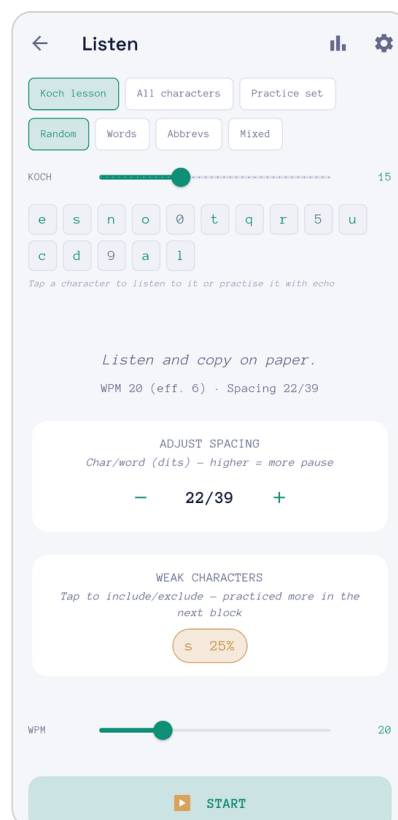
Usually you don't need to raise the lesson by hand. The block flow suggests the next character once you are ready (see When the next Koch character comes).

4.1.2 Practice set

With the **Practice set** character set, an input field appears right on the start view. Enter the characters you want to practise, for example QXZJ. Upper/lower case and spaces don't matter, and duplicate characters are ignored. The app shows how many distinct characters it found. The practice set only offers the **Random** content.



Tapping a Koch character



Listen start view: character set, content, Koch lesson, weak characters, spacing and speed

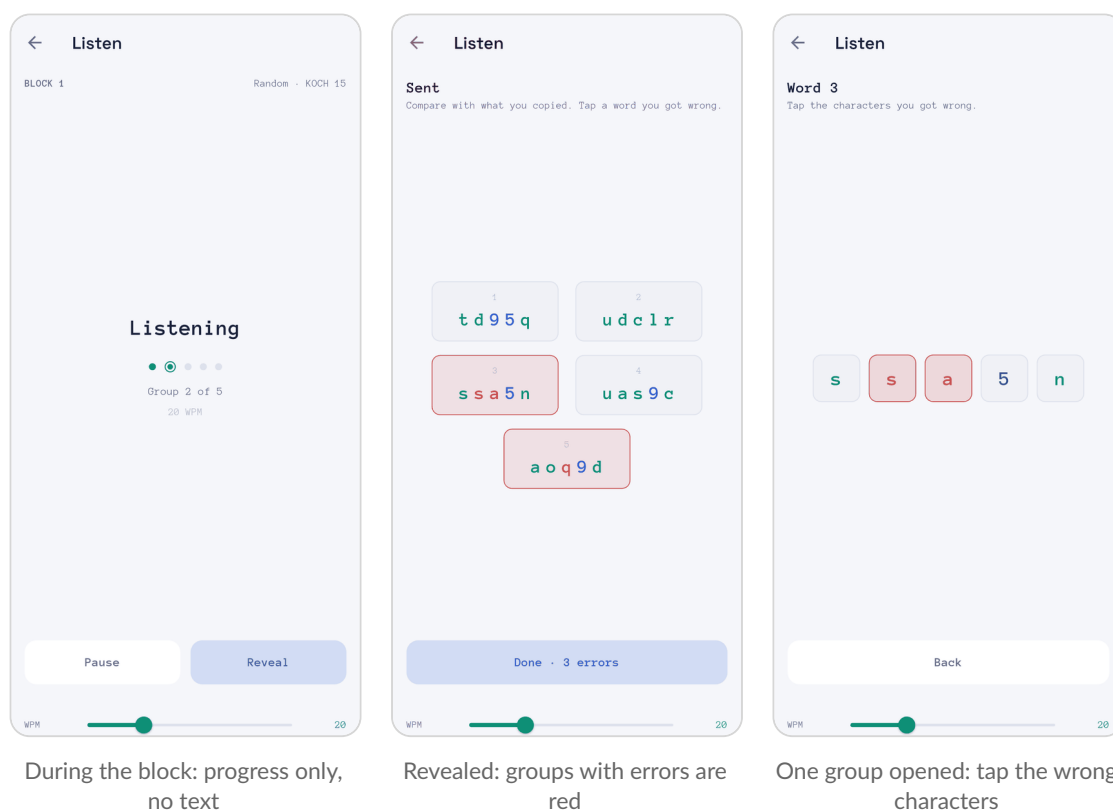
4.2 Speed

The **WPM** slider below the practice area sets the character speed (5 to 60 WPM). You set the pauses in the * sheet under **Spacing**, or directly on the start and result pages with **Adjust spacing**.

4.3 How a block runs

1. Press **Start**. After one second of "Get ready ..." the block begins.

2. The app plays the groups one after another, with the set word spacing in between. Only “Group *n* of *N*” and the speed are shown, not the text. Copy on paper.
 - **Pause** stops after the current group, and **Resume** continues.
 - **Reveal** ends the block early and shows only the groups played so far.
3. After the last group, **Sent** appears, with each group as its own tile. Compare it with what you copied.
4. **Marking errors:** tap a group where you made a mistake. It opens large, one character at a time. Tap every character you got wrong or missed, then **Back**. Tapping a character again removes the mark. The marked characters show in red in the overview.
5. **Done • *n* errors** finishes the evaluation. The results are saved now, and the result page appears.



You set how many groups a block has in the * sheet under **Word selection** → **Words per block**.

4.3.1 Stop after each group

With **Flow** → **Stop after each group** (in the * sheet), the app waits after every group:

- **Dit** (left paddle) or the **REPEAT** button plays the same group again.
- **Dah** (right paddle) or **NEXT** plays the next group.

This matches “Stop<Next>Rep” on the Morserino. It’s handy at the start, when you want to hear a group several times.

4.4 The result page

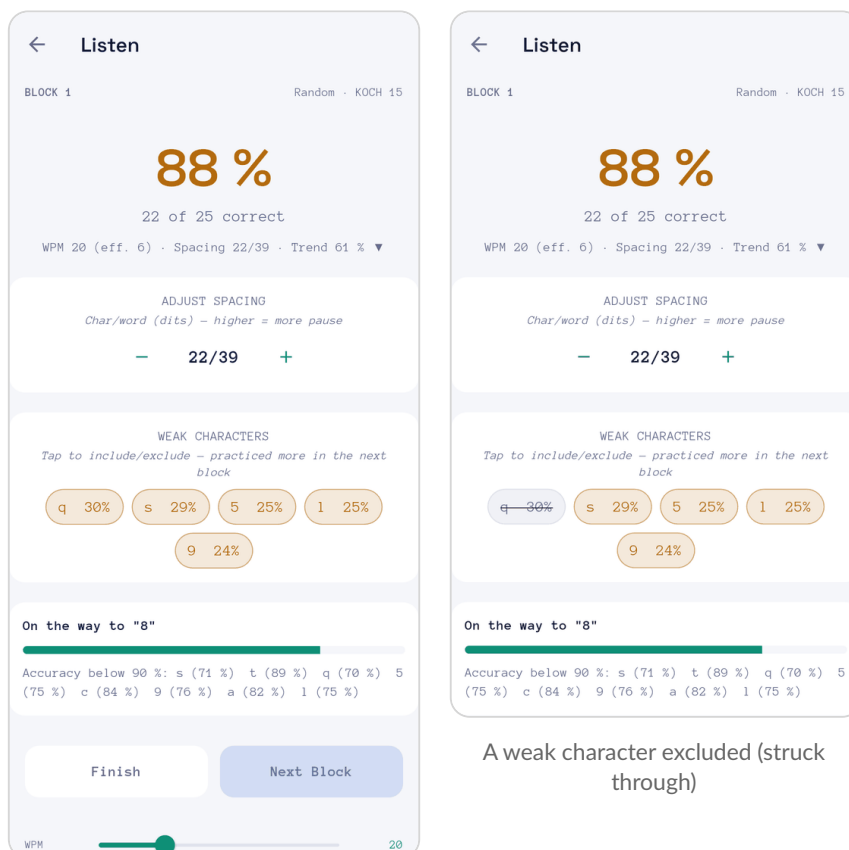
From top to bottom, the result page shows:

- **Accuracy** in percent, meaning the share of characters copied correctly, plus “x of y correct”. It is green from 90 %, yellow from 70 % and red below that.
- The **status line**: speed, effective speed, spacing and, from the sixth block on, the trend. These values already apply to the **next** block, including the ticked suggestions.
- **Adjust spacing**: – and + change Interchar Spc and InterWord Spc together by 1 dit each. This takes effect immediately and doesn't depend on the suggestions.
- **Weak characters**: the characters that cause you the most errors over time, with their error rate (see Weak characters).
- **On the way to “X”** (Koch lesson only): what you still need before the next character (see Progress card).
- **Suggestions**: what the app recommends for the next block (see Accepting, rejecting, adjusting suggestions).

At the bottom there are two buttons:

- **Next Block** applies the ticked suggestions and starts the next block right away.
- **Finish** also applies the ticked suggestions but doesn't start a new block.

How the app arrives at its suggestions is explained in the chapter Adaptive mode.



Result page with accuracy, spacing, weak characters and progress card

4.5 Settings in the ⚙ sheet (Listen)

The ⚙ sheet of **Listen** has the following sections. Values in **bold** are the defaults.

Listening

Koch Sequence
Applies to every training that uses the Koch method (Echo and Adaptive too).

Sequence

K32 LCWO CW Academy
LICW **Custom**

Custom Characters (order = learning order)

esno0tqr5ucd9al8ix1myj7h4gvkfz3b.6/w2p
39 unique characters detected

Practice Set
Custom character selection for CW-Gen mode "Practice Set" and Boost

Characters

QY
2 unique characters detected

Boost Practice

Off Moderate Strong
Draw Practice Set characters more often in Random Characters exercises

Spacing
Spacing in dit lengths, as on the Morserino (normal = 3 / 7)

Draw Practice Set characters more often in Random Characters exercises

Spacing
Spacing in dit lengths, as on the Morserino (normal = 3 / 7)

Interchar Spc 22 dits · 1.3 s @ 20 WPM

InterWord Spc 39 dits · 2.3 s @ 20 WPM

Word selection
Applies to: Koch lesson · Random

Group Length (random) 5

Words per block 5

Flow
After each group the app waits. Dit (left paddle) repeats the group, dah (right paddle) plays the next. Like "Stop<Next>Rep" on the Morserino.

Stop after each group

Adaptive Mode
Thresholds controlling how Adaptive Copy adjusts tempo/spacing and when the next Koch character unlocks.

Success Threshold Low/High 70% / 90%

⚙ sheet: Koch Sequence and Practice Set

⚙ sheet: Spacing, Word selection, Flow

4.5.1 Koch Sequence

Only shown when the **Koch lesson** character set is selected. This setting applies to **all** trainings and games that use the Koch method. For a description see Koch Sequence.

4.5.2 Practice Set

Setting	Meaning	Values
Characters	The practice set characters. This is the same field as on the start view with the Practice set character set	any characters
Boost Practice	Draws practice set characters more often in random groups (see below)	Off / Moderate / Strong

This is how **Boost Practice** works. For each character of a random group, the app draws up to 3 times (Moderate) or 8 times (Strong) until it gets a character from the practice set. If that doesn't happen, the last character drawn stays. The practice set characters come up more often, but the others don't disappear.

In **Listen**, the automatic boost of weak characters takes over this job during a block. The **Boost Practice** setting therefore mainly has an effect in **Send** (with **All characters • Random**).

4.5.3 Spacing

Setting	Meaning	Values
Interchar Spc	Pause between characters, in dits	3–45 (28)
InterWord Spc	Pause between groups/words, in dits	6–105 (40)

InterWord Spc can never be smaller than Interchar Spc. If you move Interchar Spc past it, InterWord Spc is pulled along.

4.5.4 Word selection

Only the settings that fit the chosen content are shown. The line “Applies to: ...” tells you which combination you are setting up.

Setting	Meaning	Values
Random Groups	Only with All characters • Random : which character groups are drawn from	All Chars / Alpha / Numerals / Interpunct. / Pro Signs / Alpha + Num / Num+Interp. / Interp+ProSn / Alph+Num+Int / Num+Int+ProS
Group Length	Characters per random group	2–8 (5)
Max Word Length	Only words up to this length (with Words and Mixed)	all , 1–8
Max Abbreviation Length	Only abbreviations up to this length (with Abbrevs and Mixed)	all , 2–6
Words per block	Number of groups/words in a block	1–50 (10)

4.5.5 Flow

Setting	Meaning	Values
Stop after each group	Wait after each group: dit = repeat, dah = next	Off / On

4.5.6 Adaptive Mode

These are the thresholds the app bases its suggestions on. They apply to **Listen** and **Send** together and are described in detail in Adaptive mode settings.

InterWord Spc 39 dits · 2.3 s @ 20 WPM

Word selection
Applies to: Koch lesson · Random

Group Length (random) 5

Words per block 5

Flow
After each group the app waits. Dit (left paddle) repeats the group, dah (right paddle) plays the next. Like "Stop:Next:Rep" on the Morserino.

Stop after each group ☐

Adaptive Mode
Thresholds controlling how Adaptive Copy adjusts tempo/spacing and when the next Koch character unlocks.

Success Threshold Low/High 70% / 90%

EMA Smoothing 30%

Occurrences for Unlock 20

* sheet: Adaptive Mode

5 Send: Echo Trainer

Send is the Echo Trainer. The app plays a word or group, and you key it back with the paddle. That can be the touch paddle or a real paddle (see Paddle and Morse key). If you get it right, the next word comes. If not, it is repeated.

Send has its **own profile**, independent of Listen. It keeps its own Koch lesson, speed, spacing, practice set and character statistics. What you mix up when listening isn't necessarily what you get wrong when sending.

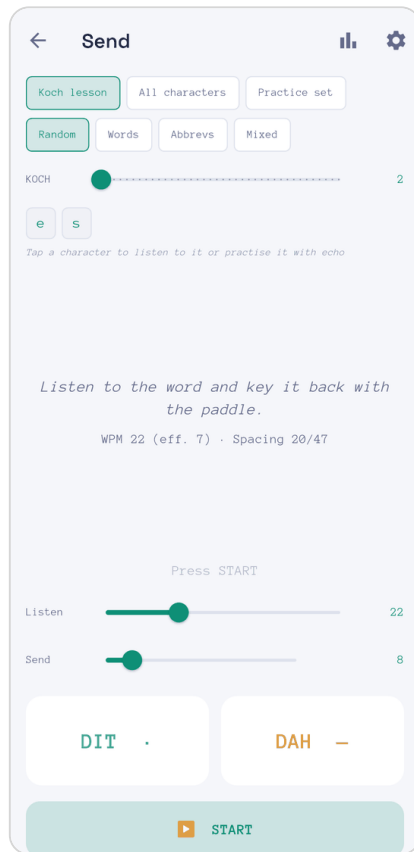
5.1 Start view

At the top you choose the **character set** and **content** as in Listen (see Choosing the character set and content). With the Koch lesson you also choose the lesson. Tapping a character works as in Listen.

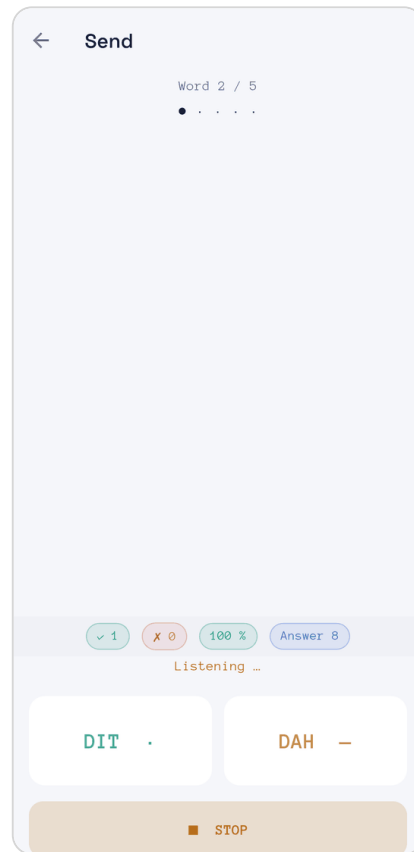
Below that, two sliders set the speeds:

- **Listen** is the speed at which the word is played to you.
- **Send** is the highest speed at which your answer is expected (see Answer speed). "same as prompt" means the same speed.

At the bottom are the paddles or the key, and **Start**.



Send start view



While answering: prompt (Echo Prompt = Both), attempt and speed

5.2 How a word runs

1. After **Start** the app waits 2 seconds, then plays the first word.
2. **Your answer:** key the word back. Your sidetone is shifted by half a tone so you can tell prompt and answer apart. This is the **Tone Shift** setting.
3. **When you have to start:** from the end of the prompt you have about 1.4 seconds, plus one character pause, plus a third of the word pause, plus the **Think Time** (default 8 s) to **begin** your answer. If nothing comes by then, the word counts as wrong.
4. **When the answer ends:** once you have started, the think time no longer applies. The answer is complete as soon as you leave a 7-dit word pause, at answer speed.
5. **Correcting:** key <err> (eight dits) or four e in a row to clear your answer so far and start again. The exception is when the word itself continues with another e at that point. Then the e counts as a normal character.
6. **Grading:**
 - **✓ Correct**, optionally with a confirmation tone. The next word comes after a little more than a second.
 - **✗ Wrong:** the word is played again. "Attempt *n* of *max*" shows which attempt this is. Once all **Repeats** are used up, the app shows the correct word for two seconds and moves on.

Whether the prompt is played, shown or both is set with **Echo Prompt** (see below).

5.3 Answer speed

On the Morserino this setting is called “Echo Speed Max”. It limits the speed at which **your answer** is expected. The prompt still plays at the listening speed.

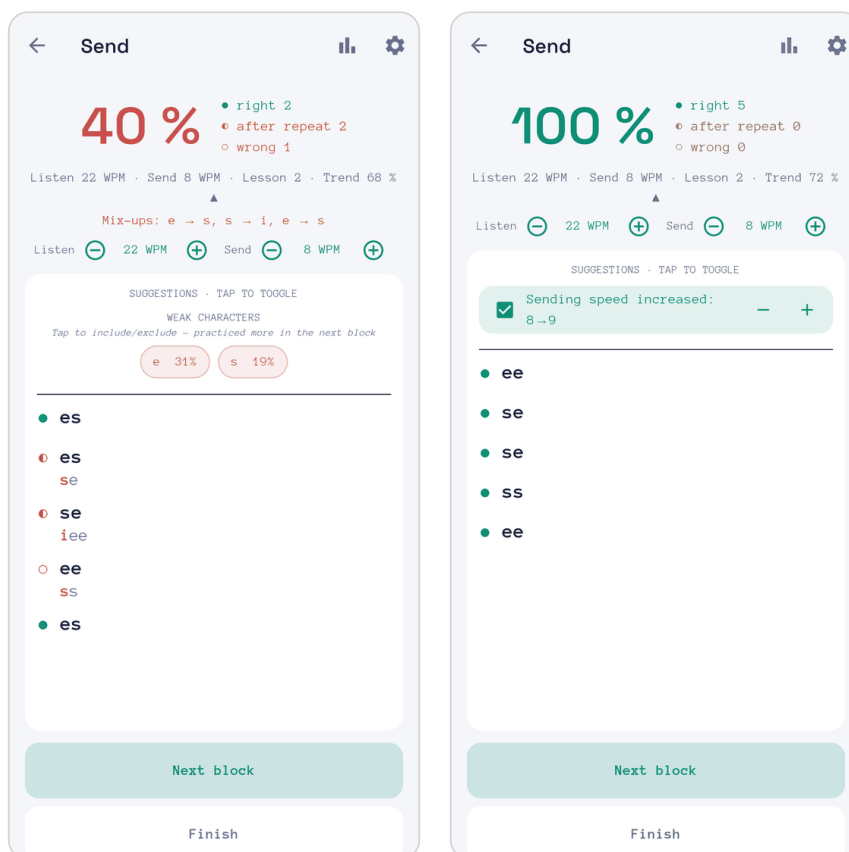
Example: Listen at 25 WPM and Send at 18 WPM means you hear fast but may answer more slowly. The answer speed sets how fast the keyer produces your dits and dahs and how long a word pause has to be. At 0 (“same as prompt”), the listening speed applies to the answer too.

5.4 The result page

After the last word of a block the result page appears:

- **Accuracy** is the share of words correct on the **first attempt**. The colours are the same as in Listen.
- Next to it is the breakdown:
 - ● **right**: correct on the first attempt.
 - ◐ **after repeat**: correct only after a repeat.
 - ○ **wrong**: not made even after all repeats.
- The **status line** shows listening speed, answer speed (if capped), lesson and the trend.
- **Mix-ups** lists which characters you confused in this block, as “target → given”, for example p → w. A — means nothing came at that position.
- The **Listen** and **Send** speed controls. A change here replaces the matching suggestion.
- **Suggestions** and **Weak characters** (see Adaptive mode).
- The list of all words in the block. The target word is shown in bold, with your **first** attempt below it and the first wrong character highlighted in red. A — means nothing came at that position.

Next block applies the ticked suggestions, boosts the ticked weak characters in the next block, and starts it. **Finish** applies the ticked suggestions without a boost and returns to the start view.



Result page: breakdown, mix-ups, weak characters, first attempts

A suggestion (here: raise the answer speed), ticked

5.5 Settings in the ⚙ sheet (Send)

The **Koch Sequence**, **Practice Set**, **Spacing** and **Word selection** sections are the same as in Listen (see Settings in the ⚙ sheet (Listen)), but they apply to the Send profile. There are two differences:

- **Spacing** only affects the word **played to you**, not your answer. Longer spacing does give you more time to begin your answer, though. Your answer is always closed with a 7-dit word pause.
- **Word selection** also uses the call sign settings from the global settings (see Call Signs).

In addition there is the **Echo Trainer** section:

Setting	Meaning	Values
Think Time	Extra time to begin your answer	1–20 s (8 s)
Repeats	How often a wrongly answered word is played again before the app reveals it. “Forever” repeats until you get it right	0–6 (3), Forever
Echo Prompt	How the prompt is given. Sound means you only hear it. Display means you only read it, with no audio. Both means you hear it and then read it once it has played	Sound / Display / Both

Setting	Meaning	Values
Answer speed (max)	Highest answer speed, see Answer speed	same as prompt, 5–50 WPM
Tone Shift (Echo)	Your sidetone while answering is half a tone above or below the prompt	No Shift / Up ½ / Down ½
Confirmation Tone	Short tone after grading: high for right, low for wrong	Off / On

Echo Prompt = Display is a good way to get from written text to sending, for example to drill new characters.

5.6 Practising a single character

In Listen or Send, tap a Koch character and choose **Practise with echo**. This opens a drill for that one character. It plays over and over, and you key it back. The drill has no blocks and no end, and it isn't counted in the statistics. If you don't answer, the character simply repeats. **Back** leaves the drill.

This corresponds to “Learn New Chr” or “Preview Char” on the Morserino.

Word selection
Applies to: Koch lesson - Random

Group Length (random) 2

Words per block 5

Echo Trainer

Think Time 5s

Repeats 2 x

Echo Prompt
Sound Display Both

Answer speed (max) 8 WPM

Your answer is expected at this speed at most. The prompt still plays at the normal speed.

Tone Shift (Echo)
No Shift Up ½ Down ½

Confirmation Tone ☒

Send * sheet: the Echo Trainer section

6 Adaptive mode

Listen and Send always run in **blocks**. After each block the app evaluates how it went and gives you **suggestions**. It may suggest unlocking the next Koch character, making the pauses shorter or longer, or raising the speed. **None of this happens behind your back**. Every suggestion appears on the result page, and you decide whether to accept it.

This chapter explains exactly how the app calculates. You don't need it to practise, but it helps you understand the suggestions and set the thresholds deliberately.

6.1 What the app tracks

6.1.1 Per character

The app keeps statistics for every character, separately for **Listen** and **Send**:

- **attempts**: how often the character was graded,
- **errors**: how many of those were wrong,
- **moving error rate**: the basis for all decisions.

The **moving error rate** is an exponential moving average (EMA). On every attempt:

new rate = $0.2 \times (1 \text{ on error, else } 0) + 0.8 \times \text{old rate}$

Each new attempt counts 20 % and the past counts 80 %. Recent results weigh more than old ones, but a single slip doesn't upset everything. For example, a character at 0 % error rate that gets one error is at 20 %. It then takes 3 correct attempts to get back below 12 %, and 8 correct attempts to get below 5 %.

A character's **accuracy** is 100 % minus its moving error rate.

What counts as an attempt?

- **Listen**: every character played in a block. It is correct unless you marked it as an error.
- **Send**: only the **first** attempt of each word. The characters before the first error count as correct, and the first wrong character counts as an error. The characters after it aren't counted, because it's unclear whether you heard them correctly. Repeats of the same word don't count.

6.1.2 Per block

Every block has a **block rate**:

- **Listen**: characters copied correctly / all characters.
- **Send**: words correct on the first attempt / all words.

From the block rates the app forms another moving average, the **block EMA**:

$$\text{block EMA} = \alpha \times \text{block rate} + (1 - \alpha) \times \text{previous block EMA}$$

α is the **EMA Smoothing** setting (default 30 %). The block EMA starts at 100 %. It is kept across sessions, separately for Listen and Send.

6.2 The two thresholds

With **Success Threshold Low/High** (default 70 % / 90 %) the app divides the block EMA into three ranges:

Block EMA	Meaning	Suggestion
90 % and above (high)	Going well	After 2 blocks in a row in this range: shorten the pauses, or raise the speed if the pauses are already normal
70 % to below 90 %	About right	Change nothing
below 70 % (low)	Too hard	Immediately: lengthen the pauses

The high threshold is also the accuracy every character has to reach before the next Koch character unlocks.

6.3 When the next Koch character comes

The next Koch character is suggested when **every** character of the current lesson meets two conditions. That means all of them, not just the most recently learned one:

1. at least **20 attempts** (the **Occurrences for Unlock** setting), and
2. an **accuracy of at least 90 %** (the high success threshold).

Two points matter here:

- The condition applies to each character individually. A single weak or rarely practised character holds up the unlock. The progress card and the character statistics show you which one it is.
- Because accuracy is a moving value, it isn't enough to have got a character right 20 times at some point. Your **recent** attempts have to be good.

The unlock appears as a highlighted suggestion with a star: **New character unlocked: "X"**. The  icon next to it plays the new character twice, without leaving the result page. If you untick it, you stay in the current lesson.

6.4 Pauses and speed

The app always changes the **pauses** first, and only then the speed:

- **Shorten:** Interchar Spc and InterWord Spc each get 1 dit shorter, down to the normal 3 / 7 dits.
- **Raise the speed:** only once the pauses are already at 3 / 7 dits does the app suggest **+1 WPM**.
- **Lengthen:** Interchar Spc and InterWord Spc each get 1 dit longer, but never longer than at the start of the session. In Listen that is when you opened the training. In Send it is the value from the * sheet.

While you are still working through the Koch sequence, the app neither shortens the pauses nor raises the speed. You should be able to concentrate on new characters without everything getting faster at the same time. Lengthening is always possible. Shortening and raising the speed only start once one of these is true:

- all characters of the Koch sequence are unlocked, or
- you practise with **All characters** or a **Practice set**.

The block in which a new character is added also never shortens the pauses or raises the speed.

You can step in yourself at any time, independently of these rules. In Listen use **Adjust spacing**, and in Send use the speed controls.

6.4.1 Answer speed in Send

Send has one extra suggestion: **Sending speed increased** (+1 WPM). It only appears if you have set an answer speed **below** the listening speed and the block reached at least the high threshold. It starts **unticked**, because the answer speed is a deliberate choice.

6.5 Accepting, rejecting, adjusting suggestions

Each suggestion is a row on the result page:

- The **checkbox** on the left accepts or rejects it. Most suggestions start ticked.
- – and + adjust the size. The speed goes up to at most 5 WPM above the current one. The pauses can be anywhere between 3 / 7 dits and the value at the start of the session.
- The status line immediately shows the values for the next block.

Nothing is applied until you leave the result page with **Next Block** or **Finish**.

6.6 Weak characters

A character counts as **weak** if both of these are true:

- it has at least **8 attempts**, and
- its moving error rate is at least **12 %**.

At most the **5** weakest are shown, worst first, with their error rate. Because the rate is tracked over time, a character stays weak until you get it right reliably again. This can take several blocks and sessions.

Tap a weak character to exclude it from the boost (it appears struck through), or to include it again.

- **Listen:** the weak characters already appear on the start view and on every result page. In the next block they are boosted at the *Moderate* level, with up to 3 draws per character (see Practice Set). This only applies to random groups (Koch lesson or All characters • Random), not to words.
- **Send:** the weak characters appear on the result page when you practise **Koch lesson • Random**. With **Next block**, the ticked characters come up twice as often in the next block. The boost lasts exactly one block.

6.6.1 Weighting in Send

In **Send • Koch lesson • Random** the app also draws the characters by weight. On the Morserino this is called “Adaptive Random”. Every character has a weight between 1 and 20, and the higher it is, the more often the character comes up. After the first attempt of each word:

- **Word fully correct:** each character of the word gets –1.
- **Error:** the first wrong character gets +4, and its neighbours in the word get +2 each (if they are different characters).

Characters you get wrong come up more often quickly. They only become rarer again once you key them correctly several times. That isn’t always pleasant, but it’s very effective.

6.7 Progress card “On the way to X”

On the **Listen** result page with the Koch lesson, a card shows what is still missing before the next character:

- The **bar** shows how many of the required attempts you have made, added up over all characters that still lack attempts. If no attempts are missing any more and only accuracy is, it shows the weakest character’s accuracy relative to the threshold.
- **Still to practise** lists characters that still lack attempts, with the number missing, for example q (7).
- **Accuracy below 90 %** lists characters with enough attempts but too low an accuracy, for example y (84 %).

6.8 Trend

From the **sixth** block on, the status line shows a trend, for example “Trend 87 % ▲”:

- The number is the average block rate of the **last 5 blocks**.
- The arrow compares it with the average of the 5 blocks before. ▲ means at least 3 percentage points better, ▼ means at least 3 points worse, and ► means the same.

The trend is kept across sessions, separately for Listen and Send. The app stores the last 20 blocks for it.

6.9 Adaptive mode settings

These are in the * sheet of **Listen** under **Adaptive Mode**. They apply to Listen and Send together.

Setting	Meaning	Values
Success Threshold Low/High	Below <i>low</i> the pauses are lengthened. At <i>high</i> (2 blocks in a row) they are shortened or the speed is raised. <i>High</i> is also the accuracy each character needs for the Koch unlock	30–99 %, at least 5 points apart (70 % / 90 %)
EMA Smoothing	How strongly the last block counts in the block EMA. Higher reacts faster, but is also jumpier	5–100 % (30 %)
Occurrences for Unlock	Minimum attempts per character before the next Koch character can come	5–50 (20)

Some notes:

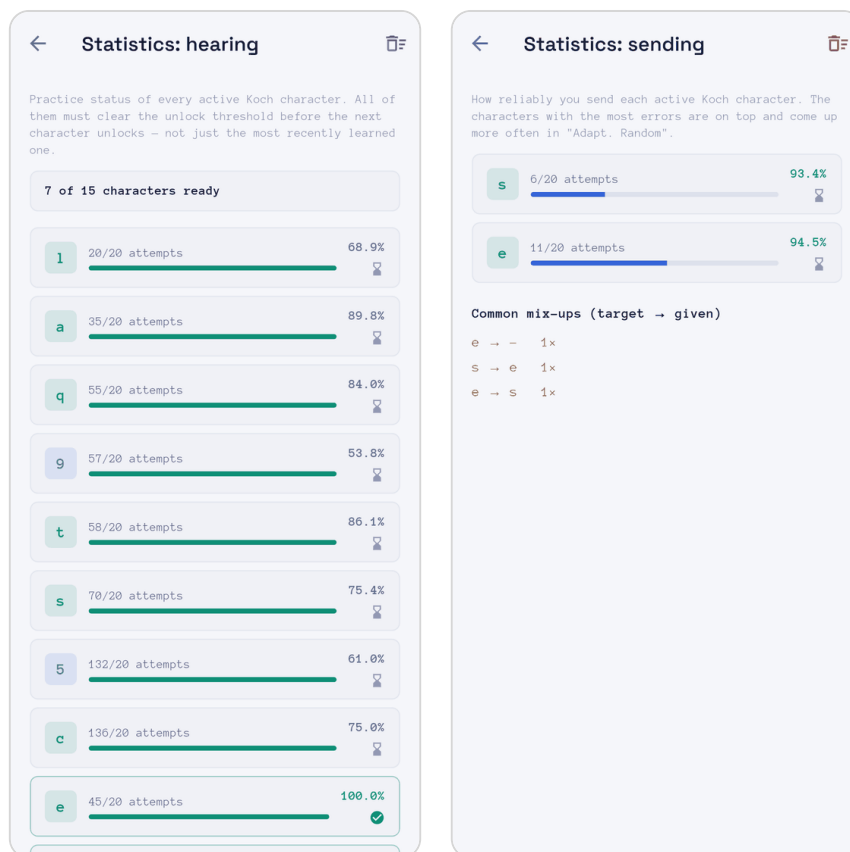
- The high threshold can't be set to 100 %. The moving error rate of a character that ever had an error never quite returns to 0. At 100 % that character would block the unlock forever.
- **EMA Smoothing 100 %** means only the last block counts.
- If unlocking goes too fast for you, raise **Occurrences for Unlock** or the high threshold. If it's too slow, lower them.

7 Character statistics

The  icon in Listen and Send opens that training's statistics. Listen and Send have separate statistics.

- **Listen** shows every active Koch character with its attempts (for example “14/20 attempts”) and its accuracy. A tick marks characters that meet the unlock condition, and an hourglass marks those that aren't there yet. The top shows “x of y characters ready”. The characters that aren't ready come first, which helps when the unlock seems stuck.
- **Send** shows the characters sorted by errors, the least reliable at the top. Below them are the **common mix-ups** across all blocks, as target → given, with a count.

The **reset** icon deletes all of this training's statistics after a confirmation: error rates, weights and mix-ups. The other training's statistics are kept. This cannot be undone.



Listen statistics: attempts, accuracy, ready

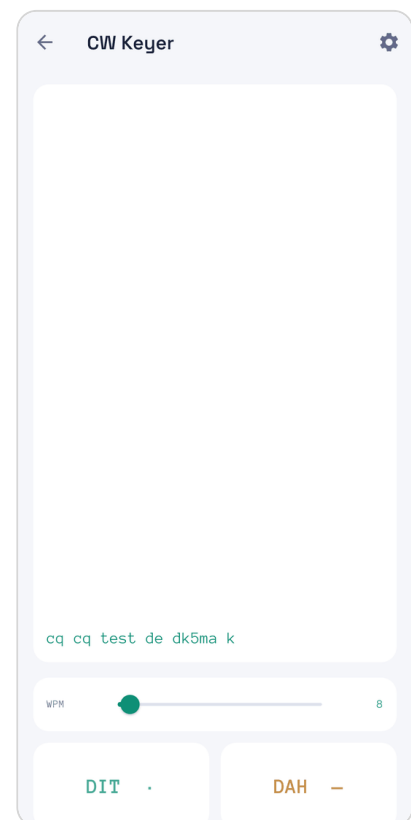
Send statistics with common mix-ups

8 CW Keyer

For free keying. What you key is heard and shown as decoded text.

- Use the **paddles** at the bottom (DIT on the left, DAH on the right), or a real paddle through an adapter (see Paddle and Morse key). In **Straight** mode a single **KEY** area appears instead.
- **WPM** sets the keyer speed, 5 to 60 WPM.
- The text runs from the bottom upwards. You can scroll back to older lines, and you change the text size with two fingers.
- The * sheet at the top right holds the **word spacing** (InterWord Spc, default 7 dits). It sets the pause after which a space is inserted. Interchar Spc has no effect when keying, as on the Morserino.

You set the keyer mode and its details in the global settings under **Keyer** (see Keyer). They apply everywhere you key: CW Keyer, Send, WiFi Trx, QSO Bot and the games.



CW Keyer with decoded text

9 CW Decoder

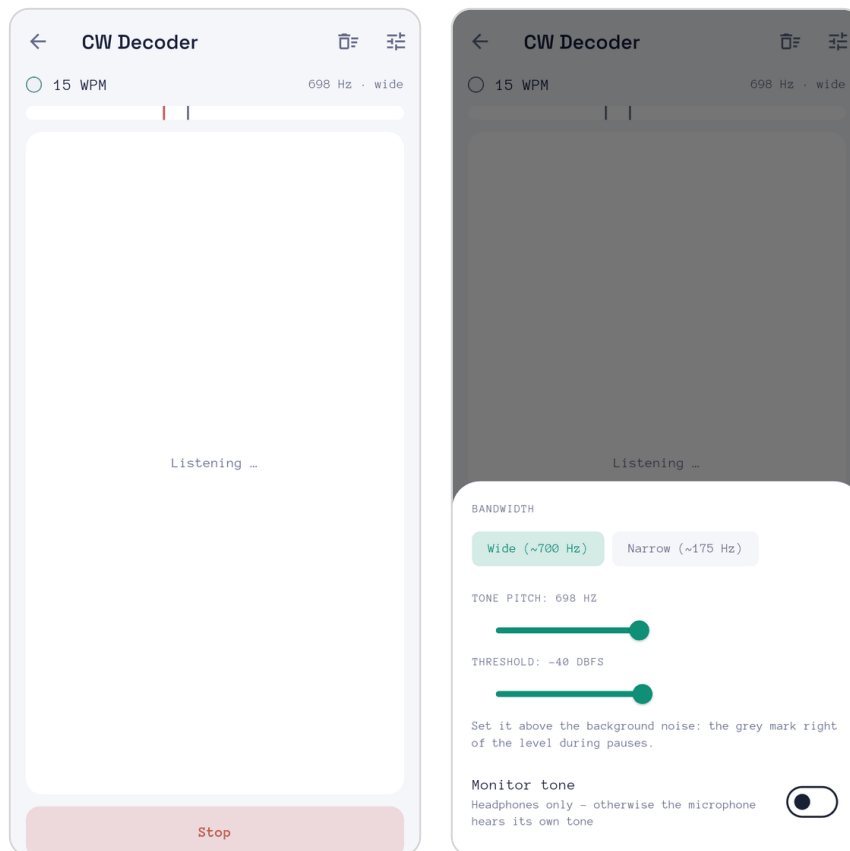
The CW Decoder listens through the **microphone** and writes down what it recognises as Morse code. The signal might come from a radio, a practice program or another Morserino.

On first start Android asks for microphone permission. Without it the decoder can't work. If you declined, allow the microphone in the Android settings under **Apps → Next CW Trainer → Permissions**.

9.1 Operation

- **Start / Stop** switches copying on and off.
- The **status line** shows the set pitch, the bandwidth and the detected speed in WPM.
- The **level meter** shows the loudness of the tone at the set pitch. A grey mark shows the noise level during pauses.
- The icons at the top right clear the text and, with the sliders icon, open the decoder settings.

The decoder adapts to the speed automatically, so there's nothing to set. It comes from the firmware. A Goertzel filter detects the tone, and timing logic tells dits, dahs and pauses apart. It keeps learning the current dit and dah length as it goes.



CW Decoder listening, with level meter

Decoder settings

9.2 Decoder settings

Setting	Meaning	Values
Bandwidth	Wide is more tolerant of a pitch that doesn't quite match. Narrow filters interference better, but the pitch must be exact	Wide (~700 Hz) / Narrow (~175 Hz)
Tone pitch	The frequency of the CW tone the decoder listens for	300–1200 Hz (698 Hz)
Threshold	Minimum loudness for a tone to count. Set it just above the grey noise mark	–65 to –10 dBFS (–40 dBFS)
Monitor tone	Plays back the detected tone cleanly at the set pitch	Off / On

Tips

- If you don't know the signal's pitch, start with **Wide** and turn the pitch until the level jumps clearly on the characters. Then switch to **Narrow**.
- Random **e** and **t** appearing in the pauses mean the **threshold** is too low.
- Only use the **monitor tone** with headphones. Otherwise the microphone hears the tone itself and confuses the decoder.

10 WiFi Trx

With WiFi Trx you send Morse over the internet or your local network with other Morserinos, apps and servers. The app uses the same protocol as the Morserino (“Morse over Packet”, MOPP, UDP port 7373). One example is the chat server **cq.morserino.info**.

The app uses the phone's existing WiFi or mobile data connection, so there are no WiFi credentials to enter.

10.1 Services

At the top you choose the **service** to connect to. The default is **cq.morserino.info**.

- **+** adds a new service, and the **pencil** edits or deletes the selected one. Each service has a **Name** and a **Server** (host name or IP address).
- An **empty server** means broadcast to the local network. This lets you exchange Morse with Morserinos on the same WiFi.
- You can only do this while not connected.

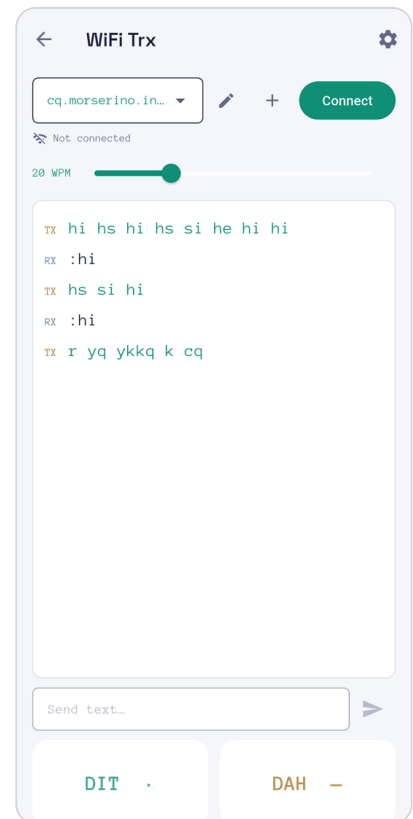
Connect connects and **Disconnect** disconnects. The line below shows the status. If the server ends the connection (**:bye**), the app disconnects too.

10.2 Sending and receiving

- **Send** with the paddles or the adapter. Each word is sent as a packet after the word pause. You can also type text into **Send text...** and send it. It goes out at the set speed.
- **WPM** sets your sending speed.
- **Receiving:** received words are played one after another at the **sender's** speed and shown in the log.
- The **log** tells received (RX) from sent (TX) text and is kept per service. **Long-press the log** to clear it after a confirmation.
- In the ***** sheet you set the **word spacing**, as in the CW Keyer. It decides after which pause your word is sent.

WiFi Trx only works while the app is in the foreground.

If your partner is behind a different router, UDP packets on port 7373 must be able to get through. With a public server such as cq.morserino.info this is normally no problem.



WiFi Trx with service, log and text field

11 QSO Bot

The QSO Bot is a simulated partner for complete, realistic radio contacts, without going on the air. The bot sends in CW, understands what you key and reacts to it. It only sounds through the app's speaker and never sends anything to the network.

11.1 Starting

1. Choose the **QSO type: SOTA/POTA, Standard or Contest**.
2. Press **Start**. The bot listens for 5 seconds:
 - If you call **CQ** (for example `cq cq de oe1abc k`), it answers you.
 - If you stay silent, it calls CQ itself and you answer.

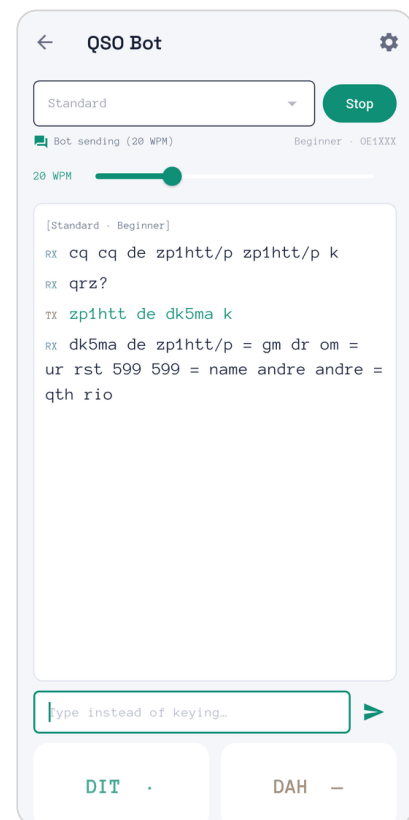
Each contact uses a new, realistic call sign for the bot.

11.2 QSO types

- **SOTA/POTA** is a summit or park activation with report and reference. If the bot calls CQ, it is the activator and you are the chaser. If you call CQ, it's the other way round. A session is a single contact.
- **Standard** is the classic QSO in three rounds. First come report, name and QTH. Then station details follow (rig, ant, wx, age), and finally the sign-off. The bot finds your details by the keywords `name`, `qth`, `rig`, `ant`, `wx` and `age`, and it remembers your name.
- **Contest** is many very short contacts in a row. The exchange depends on the **Contest type: CQ WW** (report + CQ zone) or **WPX** (report + serial number). The session ends by itself after a longer silence.

11.3 Talking to the bot

- **End each over** with `k`, `bk`, `<ar>`, `<sk>` or `73`, or simply pause. The bot won't talk over you.
- **Repeat:** `agn`, `rpt` or `?` repeats the bot's last over. `rpt rst`, `rpt call`, `rpt qth` or `rpt name` repeats just that item.
- **Bot speed:** `qrs` makes it slower and `qrq` faster. Your own speed doesn't change.
- **Correcting:** `<err>` (eight dits) or `eeee` discards your last word.
- You can send reports in cut numbers: `5nn` for 599, `t` for 0, `a` for 1, `n` for 9.
- Filler words like `de`, `r` and `ur` don't matter, because the bot picks out the information.



A QSO with the bot, answered with the text field

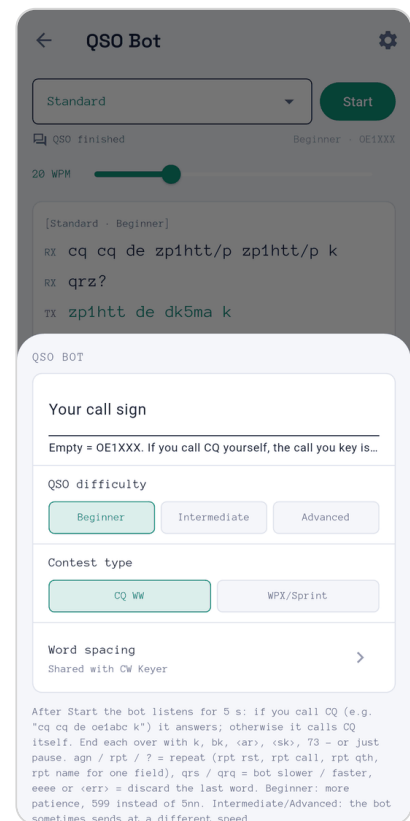
- Instead of keying, you can type text into the input field. It is treated as if keyed, but not played aloud.

11.4 QSO Bot settings

These are behind the ⚙ icon in the QSO Bot:

Setting	Meaning	Values
Your call sign	Your call sign for the bot. If you call CQ yourself, the call you key is used	empty = OE1XXX
QSO difficulty	Beginner: more patience, 599 instead of 5nn, and the bot sends at your speed. Intermediate and Advanced: the bot sometimes calls at a slightly different speed, as practice for qrs/ qrq. Advanced also keeps a tighter rhythm	Beginner / Intermediate / Advanced
Contest type	Exchange in a contest	CQ WW / WPX
Word spacing	After which pause your word counts as finished. It is shared with the CW Keyer	6–105 dits (7)

WPM sets your speed. **Long-press the log** to clear it after a confirmation.



QSO Bot settings

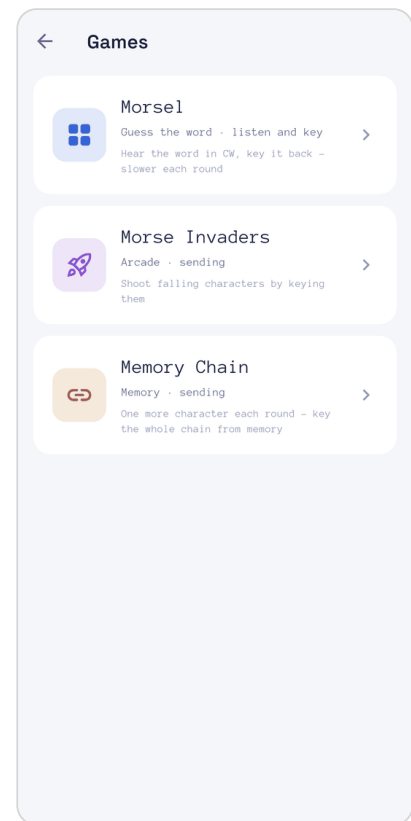
12 Games

Under **Play** → **Games** there are three games. You play them all with the paddles or the adapter. They all use the keyer settings. They take the Koch lesson from **Send**, and Morsel and Memory Chain also let you change it for the game only. Each game shows short rules before you start.

12.1 Morsel

A word puzzle like Wordle, but in CW.

- A game has **ten words**. One letter is revealed, and the word is played once in CW.
- Key the **whole word** back. A word pause submits it.
<err> (eight dits) deletes the last character.
- Colours: **green** means right, **red** means wrong, and **grey** means the revealed letter was keyed wrong.
- After each miss the word is repeated **5 WPM slower**, down to 18 WPM. After 12 seconds without input it is played again. **Skip** gives up on a word.
- **Score**: your time + 5 seconds per guess. A skipped word costs 60 seconds. Lower is better.



The games

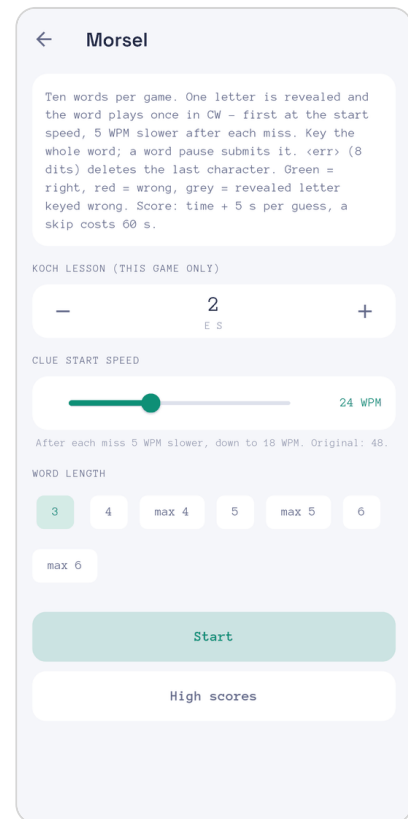
Setting	Meaning	Values
Koch lesson	For this game only. It starts at your Send lesson	2 to the end of the sequence
Word length	Exactly 3/4/5/6 letters, or at most 4/5/6	3, 4, max 4, 5, max 5, 6, max 6
Clue start speed	Speed of the first play	10–48 WPM (48 , as on the Morserino)

Longer words need a higher Koch lesson. If too few words are available, the app tells you which lesson you need.

12.2 Morse Invaders

An arcade game. Characters from your Koch lesson fall down four lanes. **Key a character** to shoot the lowest invader showing it.

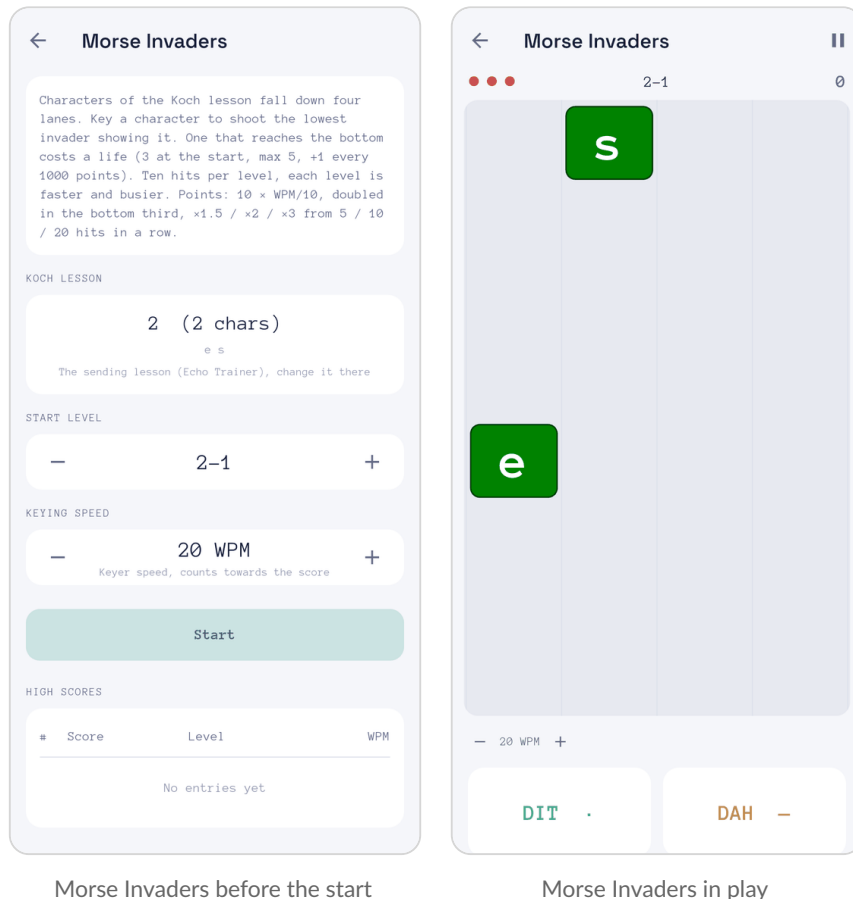
- A character that reaches the bottom costs a **life**. You start with 3 lives, can have at most 5, and gain one every 1000 points.
- After **ten hits** comes the next level. Each level is faster and busier.
- **Points:** $10 \times \text{WPM} / 10$ per hit, doubled in the bottom third. Streaks add a multiplier: $\times 1.5$ from 5, $\times 2$ from 10 and $\times 3$ from 20 hits in a row.
- **Pause** (the icon at the top right) stops the game. From there you can resume or go back to the menu. You can also change the speed with – and + during the game.



Morsel before the start

Setting	Meaning
Koch lesson	The lesson from Send (change it there)
Start level	The level you begin with
Keying speed	Keyer speed, counts towards the score

High scores are saved.



Morse Invaders before the start

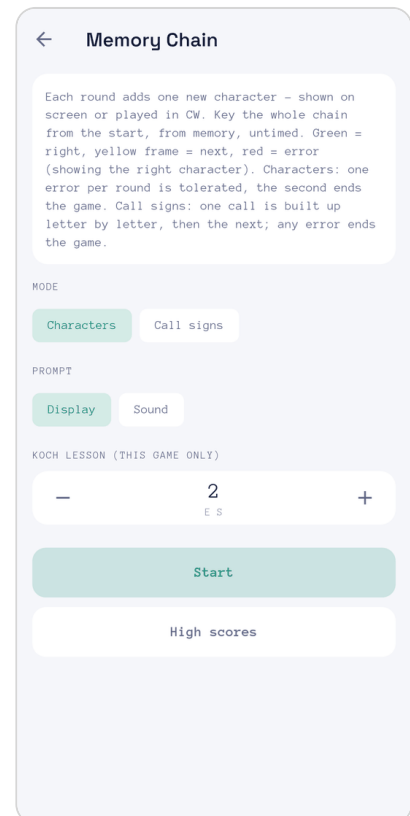
Morse Invaders in play

12.3 Memory Chain

A memory game. Each round adds **one character**, and you key the **whole chain** from the start, from memory. There is no time limit.

- **Characters mode:** random characters from your Koch lesson. One error per round is allowed, and the second ends the game.
- **Call signs mode:** a call sign is built up letter by letter, then the next one. Any error ends the game.
- **Prompt:** the new character is **shown** or **played** in CW (Sound).
- **Colours:** **green** means right, a **yellow frame** marks the next character, and **red** marks an error (with the right character).
- You set the **Koch lesson** for this game only before you start. It begins at your Send lesson. You can also change the speed with – and + during the game.

High scores are saved per mode.



Memory Chain before the start

13 Paddle and Morse key

13.1 Touch paddles

Every mode where you key shows two areas at the bottom: **DIT** (left) and **DAH** (right). In the **Straight** keyer mode there is a single **KEY** area that sounds for as long as you press it.

13.2 A real paddle or straight key

A phone has no paddle input. You need a small USB adapter that turns the paddle contacts into **key presses**:

- **vband** (hamradio.solutions/vband) is a widely used, ready-made USB adapter built exactly for this.
- **Homemade**: any small USB HID device that reports the dit and dah contacts as two different keys works. A ready-to-build example is [xiao-vband-adapter](#). It is a Seeed XIAO SAMD21 with a 3.5 mm jack that sends the same keys as the vband adapter. It plugs straight into the phone with a USB-C cable.

Older phones with micro-USB need a USB OTG adapter.

13.2.1 Learning the paddle keys

This tells the app which key your adapter sends for dit and which for dah:

1. Plug in the adapter.
2. Go to **Settings → vband Paddle → Learn Paddle Keys**.
3. When “Press Dit key ...” appears, press the **dit paddle**. When “Press Dah key ...” appears, press the **dah paddle**.
4. “Saved” confirms it. The detected keys are listed under **Dit** and **Dah**.

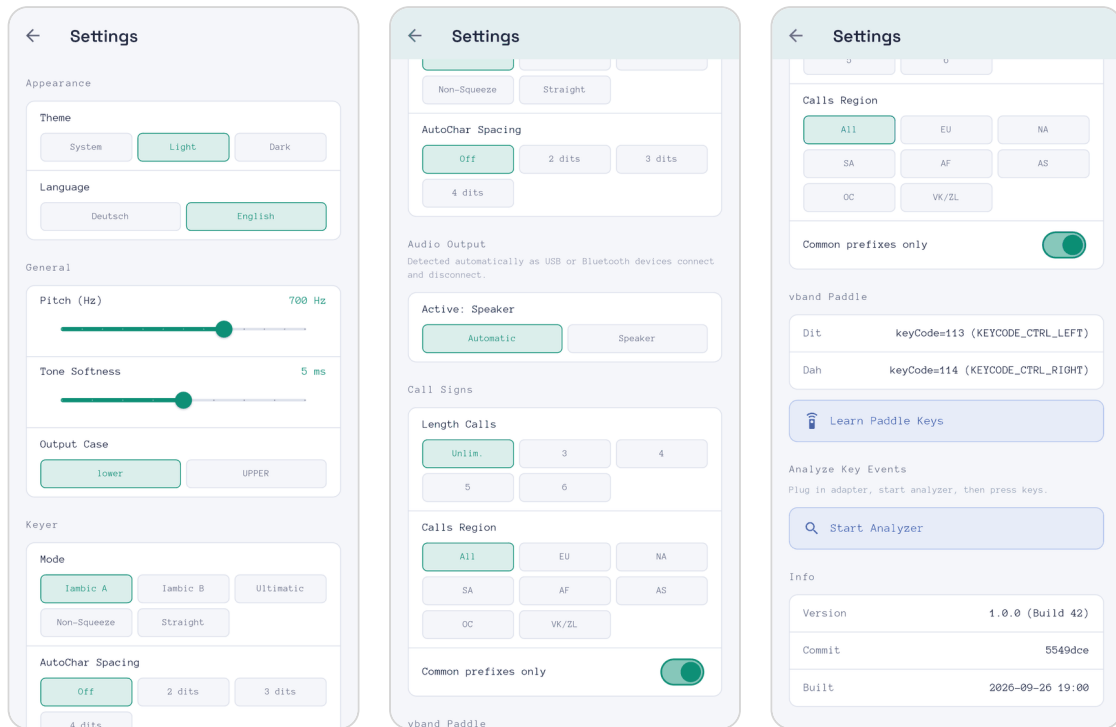
So there's no list of supported adapters. The app learns whatever keys your adapter sends.

13.2.2 Analyze Key Events

If an adapter doesn't behave as expected, use **Settings → Analyze Key Events**. Press **Start Analyzer**, then press the paddles. The app lists every key event it receives, so you can see whether the adapter sends anything and what it sends. **Stop Analyzer** ends the display.

14 Settings

Open the global settings with the gear at the top right of the home screen. They apply to the whole app. Anything that concerns only one training is in that training's * sheet.



Settings: Appearance, General, Keyer

Audio output and Call Signs

vband Paddle, key events, Info

14.1 Appearance

Setting	Meaning	Values
Theme	Light or dark look	System / Light / Dark
Language	App language	Deutsch / English

14.2 General

Setting	Meaning	Values
Pitch (Hz)	Frequency of the sidetone and of the played characters	300–900 Hz in 50 Hz steps (600 Hz)
Tone Softness	Rise and fall time of the tone. Larger values sound softer and click less, especially on short dits	1–9 ms (5 ms)
Output Case	Show characters in lower or UPPER case. Display only	lower / UPPER

14.3 Keyer

These settings apply wherever you key.

Setting	Meaning	Values
Mode	How the keyer reads the paddles (see below)	lambic A / lambic B / Ultimatic / Non-Squeeze / Straight
CurtisB Dit Timing	lambic B and Ultimatic only: from what percentage of a dit a press on the other paddle is already stored	0–100 % in steps of 5 (75 %)
CurtisB Dah Timing	The same for dahs	0–100 % in steps of 5 (45 %)
AutoChar Spacing	Enforces a minimum pause between characters so they don't run together	Off / 2 / 3 / 4 dits

The keyer modes

- **lambic A**: if you hold both paddles ("squeeze"), dits and dahs alternate. When you let go, the keyer stops after the current element.
- **lambic B**: like A, but the keyer remembers a press on the other paddle that comes during an element, and adds that element (Curtis B behaviour). The CurtisB settings control from when this applies. 0 % means during the whole element, and 100 % means practically like **lambic A**.
- **Ultimatic**: when both paddles are pressed, the **last** one pressed wins and repeats for as long as it is held.
- **Non-Squeeze**: for single-lever paddles or those switching over. Squeezing both paddles produces no alternating sequence.
- **Straight**: a straight key. The tone is on for as long as the key is pressed. With touch, a single **KEY** area appears. With an adapter, the dit contact acts as the key.

14.4 Audio output

Setting	Meaning	Values
Active	Shows where the sound is going right now	–
Output	Automatic follows whatever is currently plugged in or connected. The other options fix the output. Only outputs that are currently available are offered	Automatic / Speaker / Wired/USB / Bluetooth

14.5 Call Signs

These are the settings for random call signs in the **Callsigns** content (with All characters).

Setting	Meaning	Values
Length Calls	Maximum call sign length	Unlim. / 3 / 4 / 5 / 6
Calls Region	Only call signs from this region	All / EU / NA / SA / AF / AS / OC / VK/ZL
Common prefixes only	Only frequently heard prefixes instead of all possible ones	Off / On

Call signs follow a weighted prefix table, as on the Morserino. Frequently heard countries come up more often.

14.6 vband Paddle and Analyze Key Events

See Learning the paddle keys and Analyze Key Events.

14.7 Info: version and build

Row	Meaning
Version	Version number and build number, for example “1.0.0 (Build 42)”
Commit	The exact source code state the app was built from
Built	Date and time of the build

When you report a problem, please include these three values.

14.8 Koch Sequence

You set the Koch Sequence in the * sheet of **Listen** or **Send** when the **Koch lesson** character set is selected there. It applies to **all** trainings and games, though.

Sequence	Description
M32	The Morserino-32 order (45 characters): m k r s u a p t l o w i . n j e f 0 y v , g 5 / q 9 z h 3 8 b ? 4 2 7 c 1 d 6 x - = + @ :
LCWO	The lcwo.net order
CW Academy	The CW Academy (CWops) order
LICW	The Long Island CW Club order with an entry point (see below)
Custom	Your own order

LICW Entry Point (0–13): in the LICW course, students join a “carousel” at different points. The entry point rotates the sequence so that it starts there.

Custom: enter the characters in the order you want to learn them. Duplicates are ignored, and the app shows how many characters it found. The default is
esno0tqr5ucd9al8ix1myj7h4gvkfz3b.6/w2p?.

Prosigns are not part of the Koch sequences in the app.

15 What the app does not do (yet)

Some Morserino-32 features are missing from the app, because a phone doesn't have the hardware or Android handles it already. These are the rotary knob and buttons, the display, LoRa, ESP-NOW (and with it the multiplayer parts of the games), iCW/Ext Trx and keying a real transmitter, firmware updates and the WiFi setup page. Instead of the firmware's Practice Stats, the app has its own, more detailed character statistics.

Not done yet, but planned: the games Trailblazer, Fox Hunt, Radio Cave and Fight the Pileup, the File Player (your own text as practice content), saved settings profiles and CW Memories.

16 Troubleshooting

The app doesn't key when I press my paddle. Learn the paddle keys (see Learning the paddle keys). If nothing arrives, check with **Analyze Key Events** whether the adapter sends anything at all.

Sound comes out of the wrong device. Fix the output under **Settings → Audio output**.

When sending, I hear my tone late. This is almost always Bluetooth headphones. Use wired headphones or the speaker.

The next Koch character doesn't come. Open the  statistics in Listen. The characters without a tick are holding up the unlock. Usually it's a character with too few attempts or recent errors. See When the next Koch character comes.

The pauses never get shorter. That's intended while you work through the Koch sequence (see Pauses and speed). You can shorten them yourself at any time with **Adjust spacing**.

The decoder only writes garbage. Check pitch, bandwidth and threshold (see Decoder settings). Only turn on the monitor tone with headphones.